



Benitoite

△ Deep blue benitoite gem

Benitoite was discovered in 1906 near the San Benito River in California, from which it takes its name. It is reputed to have been found by a prospector looking for mercury and copper deposits, who came across some brilliant blue crystals he mistook for sapphires. Its best blue colour is seen through the side of its crystals, a fact that imposes a size limitation on cut stones, which seldom exceed three carats. Benitoite has exceptionally strong dispersion: its “fire” is similar to that of diamond, though it is often masked by the intensity of the stone’s colour.

Specification

Chemical name	Barium titanium silicate	Formula	$\text{BaTiSi}_3\text{O}_9$
Colours	Blue, colourless, pink	Structure	Hexagonal
Hardness	6.5	SG	3.7
RI	1.76–1.80	Lustre	Vitreous
Streak	White	Locations	USA, Belgium, Japan



Uncut benitoite | Rough | Specimens of benitoite are rarely found in rough form weighing over 5 carats – this unusually fine piece weighs nearly 7 carats.



Benitoite crystals | Rough | Although the benitoite crystals in this specimen are not gem quality, they are accompanied by calcite, which is typical of the mineral.



Fine gem | Cut | The deep colour and natural fire of this benitoite gemstone have been enhanced by the use of a cut that features a large number of facets.



Butterfly brooch | Set | Made to celebrate the California state gemstone, this brooch is set with blue and colourless benitoites, with heat-treated orange benitoites for eyes.



Heat-treated benitoite

Colourless benitoite